Calcurion Physics Worksheets

A-Level Physics

Topic: Magnetic Fields

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. State the equation for the force on a current-carrying conductor in a magnetic field. **[1]**

2. A wire of length 0.15 m carries 4.0 A perpendicular to a 0.25 T field. Calculate the force. **[2]**

3. State Fleming's left-hand rule. **[2]**

4. State the equation for the force on a moving charge in a magnetic field. **[1]**

5. An electron moves at 2.0×10^6 m/s perpendicular to a 0.10 T field. Calculate the force. ($e = 1.6 \times 10^{-19}$ C) **[3]**

6. What is the path of a charged particle moving perpendicular to a uniform magnetic field? **[1]**

7. Derive the radius of the circular path: $r = mv/BQ$. **[3]**

8. Define magnetic flux. **[1]**

9. A coil of area 0.020 m^2 is perpendicular to a 0.15 T field. Calculate the flux through the coil. **[2]**

10. State Faraday's law of electromagnetic induction. **[2]**

11. State Lenz's law. **[2]**

12. The flux through a coil of 50 turns changes from 0.020 Wb to 0.005 Wb in 0.10 s. Calculate the induced e.m.f. **[3]**

13. State the transformer equation. **[1]**

14. A transformer has 200 primary turns and 50 secondary turns. Primary voltage is 240 V. Calculate secondary voltage (ideal). **[2]**

15. What is the purpose of the soft iron core in a transformer? **[2]**

Section B – Hard (Q16–25)

16. A proton moves in a circle of radius 0.50 m in a 0.20 T field. Calculate its speed. ($m_p = 1.67 \times 10^{-27}$ kg) **[3]**

17. Explain why a charged particle moving parallel to a magnetic field experiences no force. **[2]**

18. A rectangular coil of 100 turns, area 0.01 m^2, rotates at 50 rev/s in a 0.05 T field. Calculate the peak e.m.f. **[4]**

19. Explain how a simple a.c. generator works. **[3]**

20. An ideal transformer has input 100 W. Secondary voltage 12 V and current 5.0 A. Is energy conserved? Justify. **[3]**

21. Calculate the magnetic field strength 2.0 cm from a long straight wire carrying 5.0 A. ($\mu_0 = 4\pi \times 10^{-7}$) **[3]**

22. Two parallel wires 5.0 cm apart each carry 3.0 A in the same direction. Calculate the force per unit length between them. **[3]**

23. Explain eddy currents and one way to reduce them in a transformer core. **[3]**

24. A search coil is used to measure magnetic field. Describe the method. **[3]**

25. Derive the expression for the force between two parallel current-carrying wires. **[4]**

End of Worksheet • *Show all working clearly.*